This exam has 2 pages and 7 exercises.

The result will be computed as (total number of points plus 10) divided by 10.

By participating in this exam, you underwrite the following `Statement of Integrity`:

I declare to understand that taking an online exam during this corona crisis is an emergency measure to prevent study delays as much as possible. I know that fraud control will be tightened and realize that a special appeal is being made to trust my integrity. With this statement, I promise to make this exam completely on my own, only consult those sources that are allowed explicitly, not share my solutions with other students, and make myself available for any oral clarifications regarding this exam.

1. **Semantic equivalence** $\equiv$ (9 points)
Group the following eight formulas into their semantic equivalence classes (i.e., state which formulas are semantically equivalent):

$$p \to (q \to r) \qquad (p \to q) \to r \qquad (p \to q) \leftrightarrow r \qquad (p \leftrightarrow q) \leftrightarrow r$$
$$p \leftrightarrow (q \leftrightarrow r) \qquad (p \wedge q) \to r \qquad p \oplus (q \oplus r) \qquad \neg p \vee \neg q \vee r$$

2. **Island puzzle** (12 points)
On the island of liars and truth speakers, everybody is either a liar (who always lies) or a truth speaker (who always speaks the truth). You meet islanders A and B.

A says: "If I am a liar, then B is a truth speaker."

B says: "Exactly one of us two is a liar."

Determine via a truth table for both A and B whether this islander is a truth speaker or a liar.

3. **Conjunctive normal form** (12 points)
Apply the algorithm `CNF` to turn the formula

$$\neg((p \vee q) \to r) \vee (\neg p \wedge r)$$

into CNF.

4. DPLL procedure (12 points)
Apply the DPLL procedure to the CNF

$$(p \vee q) \wedge (\neg p \vee \neg r) \wedge (\neg q \vee \neg r) \wedge (\neg p \vee r)$$

to check whether it is satisfiable. Explain in detail how the procedure comes to a result. Let the procedure branch first on p, then on q, and then on r; and let it always try the truth value true ($\top$) before false ($\bot$).

5. Sets (5+5+5 points) Consider the following set-theoretic equality

$$A \backslash (B \cap C) = (A \cap B') \cup (A \cap C').$$

a) Prove this identity using a sequence of Venn diagrams.

b) Prove it using the rules of the algebra of sets.

c) Assuming that $\#A = 5$, $\#(B' \cup C') = 5$, $\#(A \cup B' \cup C') = 7$, compute the number of elements in $A \backslash (B \cap C)$.

6. Relations (6+5+6 points) In the set $V = \{1, 2, 3, 4\}$ consider the relations

$$\begin{aligned} R &= \{\langle 1,2\rangle, \langle 1,3\rangle, \langle 1,4\rangle, \langle 2,2\rangle, \langle 2,3\rangle, \langle 2,4\rangle, \langle 3,3\rangle\}, \\ S &= \{\langle 1,1\rangle, \langle 2,3\rangle, \langle 3,4\rangle\}. \end{aligned}$$

a) Write down the matrix representation of the relation R.

b) Is the relation R transitive? Is it reflexive? (Explain your answers.)

c) Draw a Venn diagram depicting the relation $S \circ R$.

7. Ordering relations (7+6 points) Consider $A = \{1, 2\}$ and $B = \mathcal{P}(A)$, the power set of A, both equipped with their natural partial ordering: for A the relation $\leq$, and for B the relation given by set inclusion.

a) Draw the Hasse diagram of the lexicographic order of $B \times A$. Explain whether it is a total ordering.

b) Let C denote the set B with the empty set removed. List the minimal and the maximal elements of $C \times A$ equipped with the Cartesian order. Is there a largest or a smallest element of $C \times A$?